



Electrical Machines

Electrical Machines

An electrical machine is a **device which converts mechanical energy into electrical energy or vice versa**. Electrical machines also include transformers, which do not actually make conversion between mechanical and electrical form but they convert AC current from one voltage level to another voltage level.

Types of Machine: Static Machines and Dynamic Machines

- **Static Machines** – Transformers (A static machine is a **machine that performs work without any movement of its parts**).
- **Dynamic Machines** – Generator and Motors (A "dynamic electric machine" typically refers to an electric machine, such as an electric motor or generator, that operates in a dynamic or changing environment. These machines are designed to convert electrical energy into mechanical energy (in the case of motors) or vice versa (in the case of generators) in response to varying conditions or input signals)

Transformer: In the transformer, both input and output are electrical power.

Generator: In a generator, the input is mechanical power and the output is electrical power.

Motor: In a motor, the input is electrical power and output is mechanical power.



DC Generator

A DC generator is an electromechanical energy conversion device that converts mechanical power into DC electrical power through the process of electromagnetic induction.

A DC generator functions based on electromagnetic induction. This means that when there is a change in the magnetic flux linking a conductor, it induces an electromotive force (EMF) in the conductor.

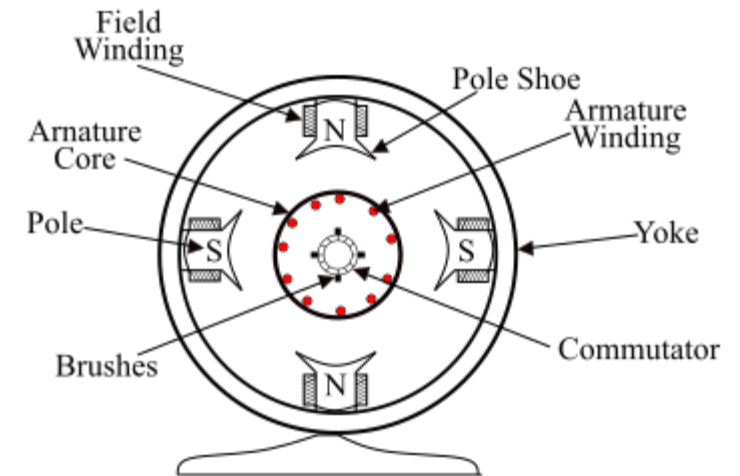
The EMF induced in the armature winding of a DC generator is alternating one and is converted into direct voltage using a commutator mounted on the shaft of the generator. The armature winding of DC Generator is placed on the rotor whereas the field winding is placed on the stator.

Construction of a DC Generator

A DC generator is an electromechanical energy conversion device that converts mechanical power into DC electrical power through the process of electromagnetic induction.

A DC generator consists of six main parts, which are as follows

- **Frame or Yoke**
- **Magnetic Field System**
- **Armature Core**
- **Armature Winding**
- **Commutator**
- **Brushes**



Construction of a DC Generator

Frame or Yoke

The outer frame of a DC generator is a hollow cylinder made up of cast steel or rolled steel is known as yoke. The yoke serves following two purposes.

- It supports the field pole core and acts as a protecting cover to the machine.
- It provides a path for the magnetic flux produced by the field winding.

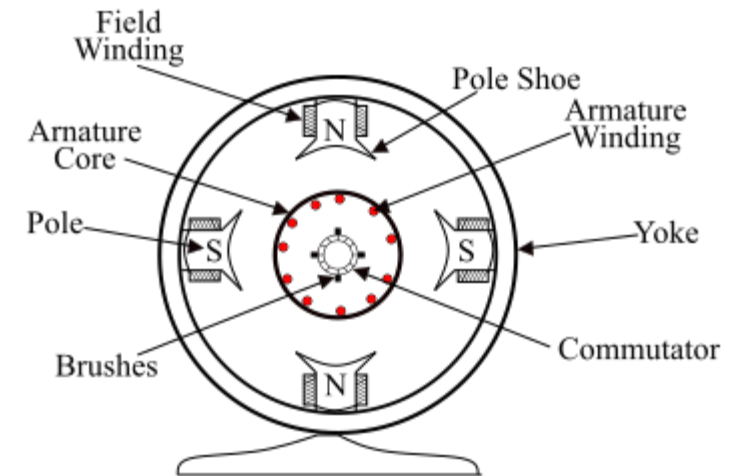
Magnetic Field System

The magnetic field system of a DC generator is the stationary part of the machine. It produces the main magnetic flux in the generator. It consists of an even number of pole cores bolted to the frame and field winding wound around the pole core. The field system of DC generator has salient poles i.e. the poles project inwards and each pole core has a pole shoe having a curved surface. The pole shoe serves two purposes

- It provides support to the field coils.
- It reduces the reluctance of magnetic circuit by increasing the cross-sectional area of it.

The pole cores are made of thin laminations of sheet steel which are insulated from each other to reduce the eddy current loss. The field coils are connected in series with one another such that when the current flows through the coils, alternate north and south poles are produced in the direction of rotation.

Dr. H. Maity



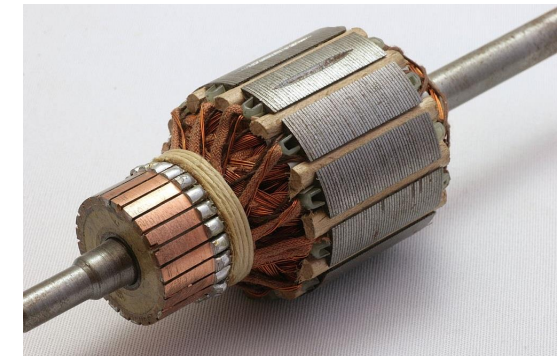
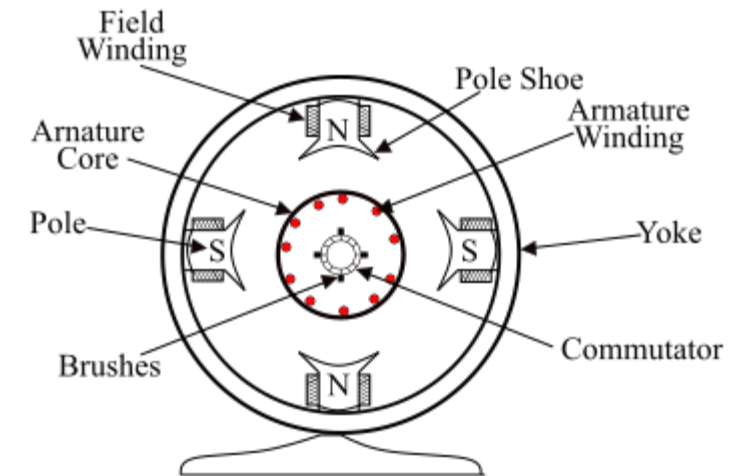
Construction of a DC Generator

Armature Core

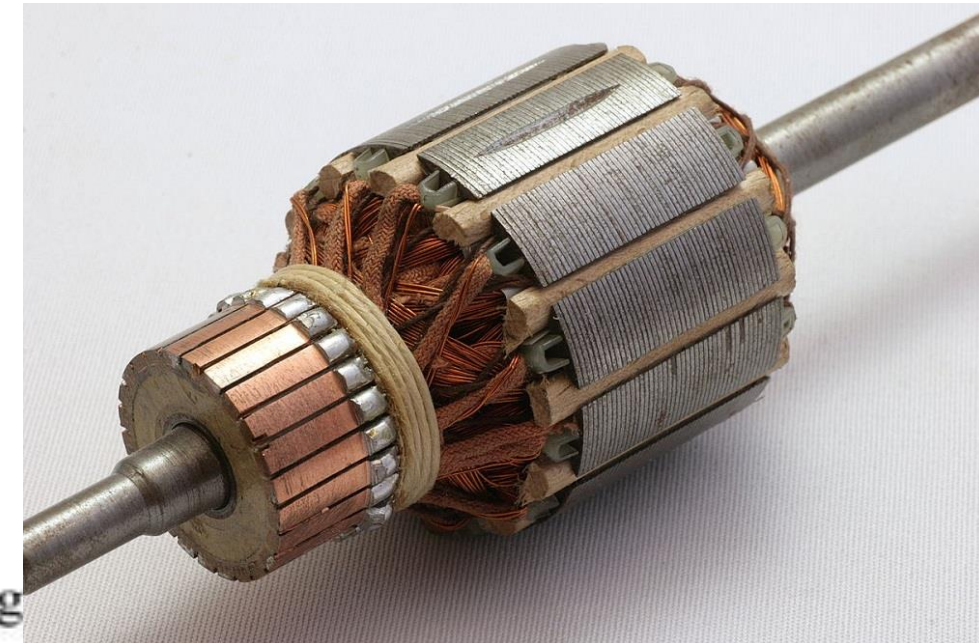
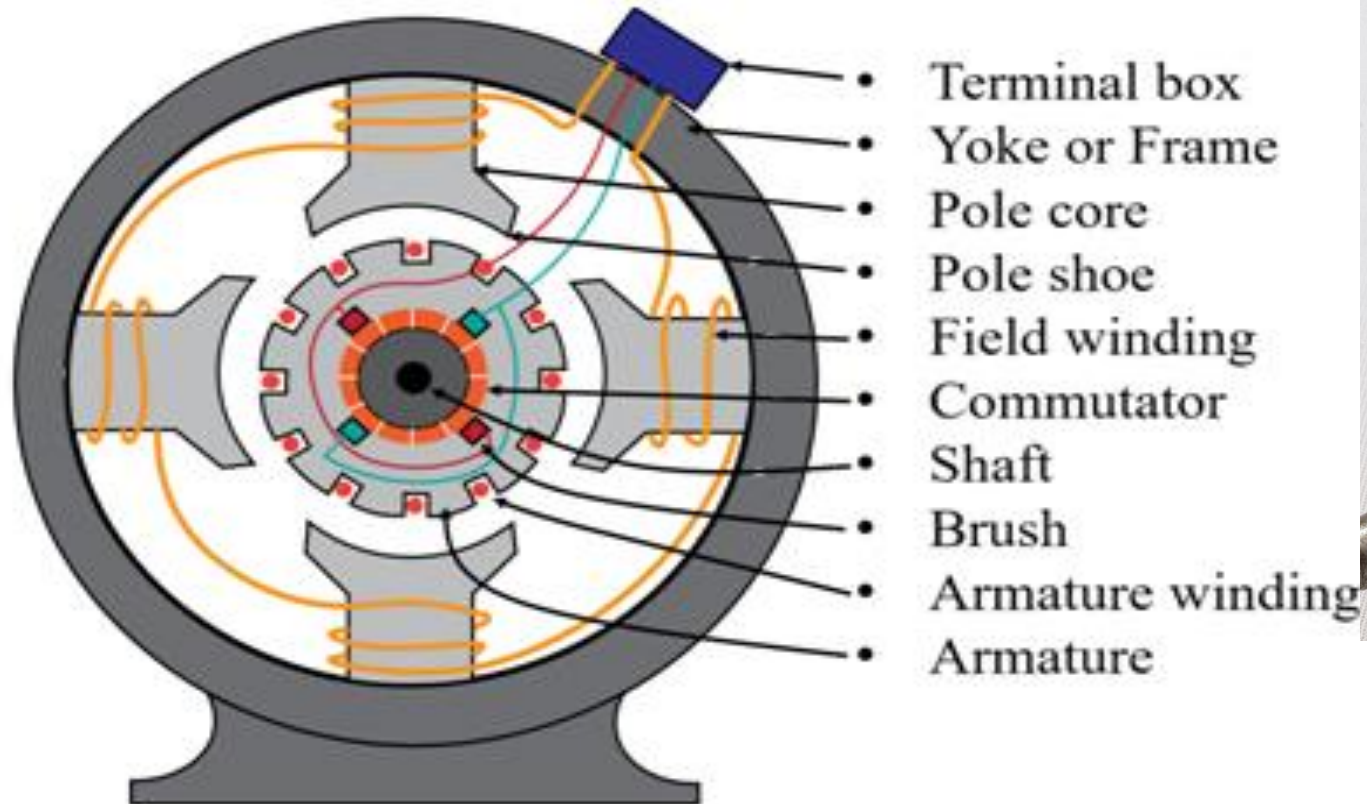
The armature core of DC generator is mounted on the shaft and rotates between the field poles. It has slots on its outer surface and the armature conductors are put in these slots. The armature core is made up of soft iron laminations which are insulated from each other and tightly clamped together. In small machines, the laminations are keyed directly to the shaft, whereas in large machines, they are mounted on a spider. The laminated armature core is used to reduce the eddy current loss.

Armature Winding

The insulated conductors are put into the slots of the armature core. The conductors are suitably connected. This connected arrangement of conductors is known as armature winding. There are two types of armature windings are used – wave winding and lap winding.



Construction of a DC Generator



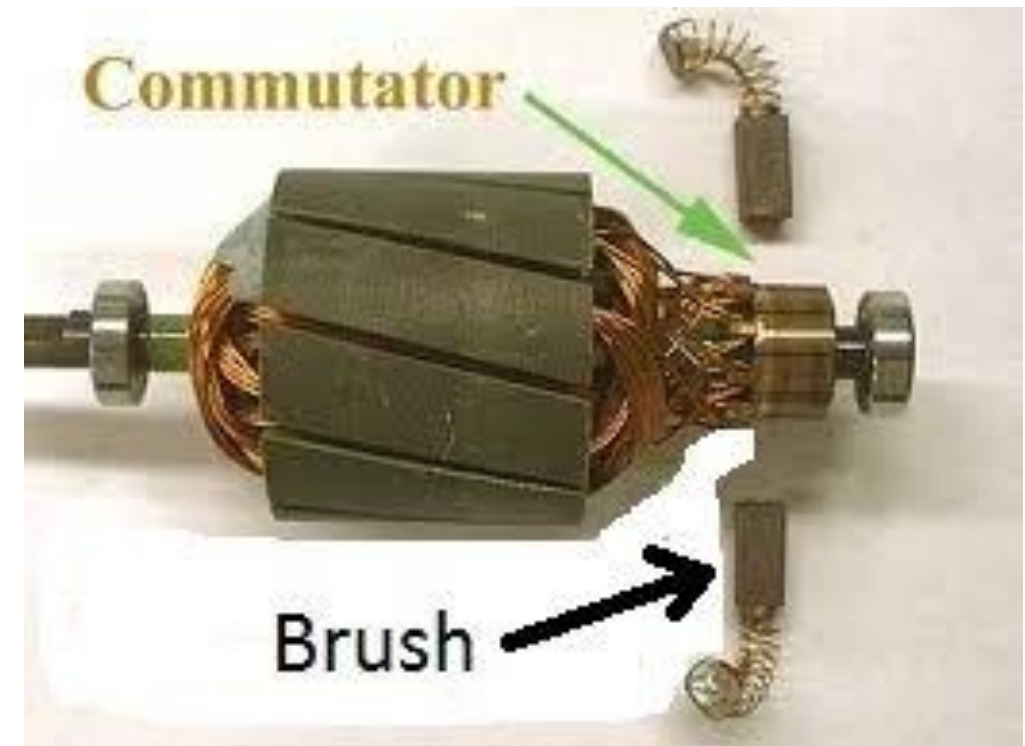
Construction of a DC Generator

Commutator

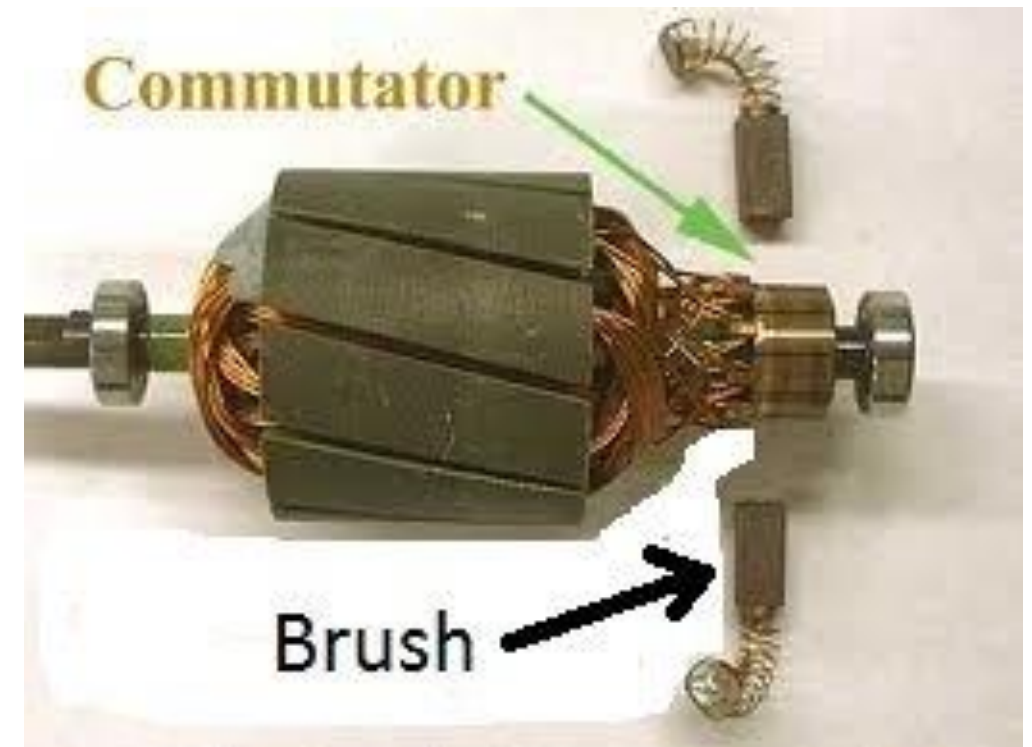
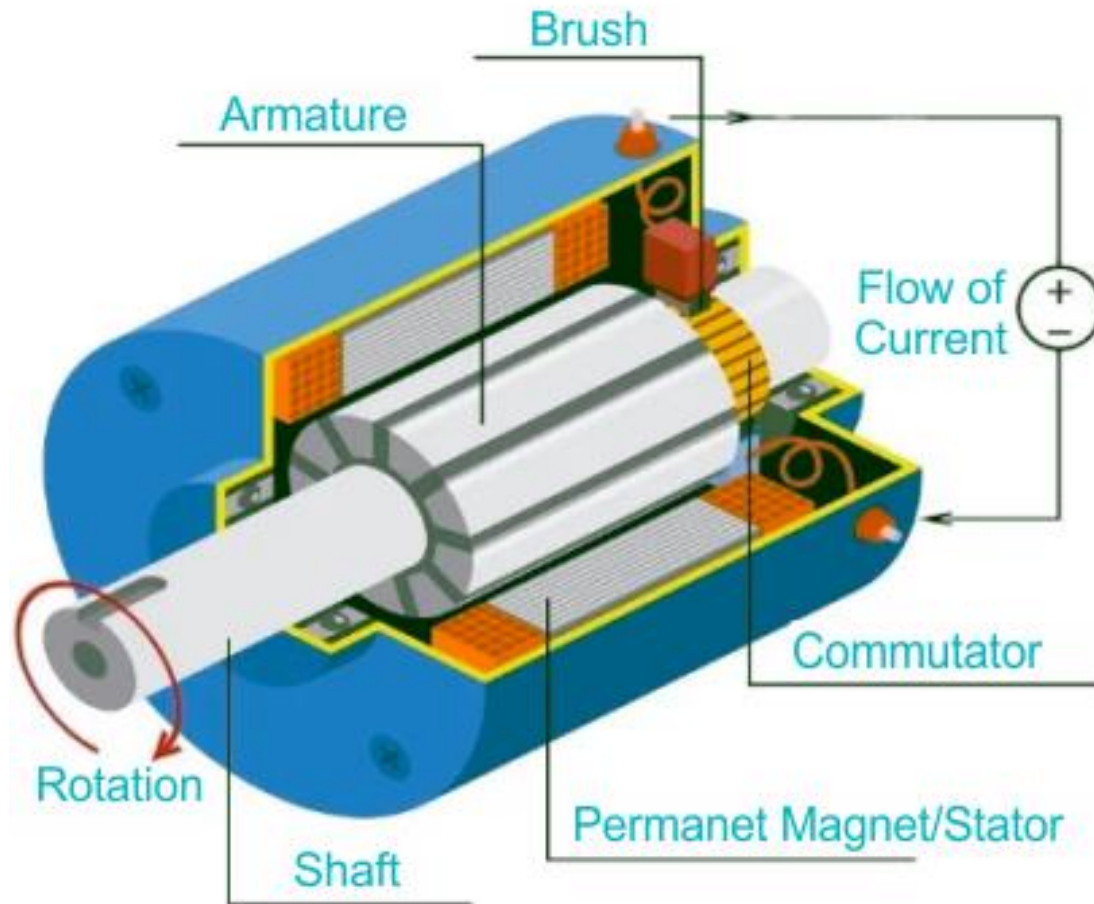
A commutator is a mechanical rectifier which converts the alternating emf generated in the armature winding into the direct voltage across the load terminals. The commutator is made of wedge-shaped copper segments insulated from each other and from the shaft by mica sheets. Each segment of commutator is connected to the ends of the armature coils.

Brushes

The brushes are mounted on the commutator and are used to collect the current from the armature winding. The brushes are made of carbon and are supported by a metal box called brush holder. The pressure exerted by the brushes on the commutator is adjusted and maintained at constant value by means of springs. The current flows from the armature winding to the external circuit through the commutator and carbon brushes.

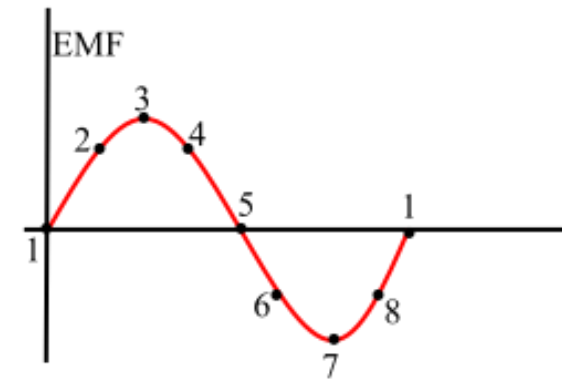
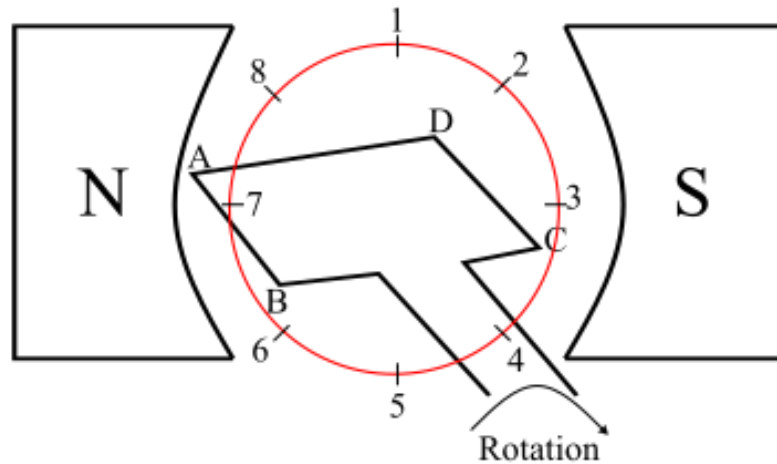
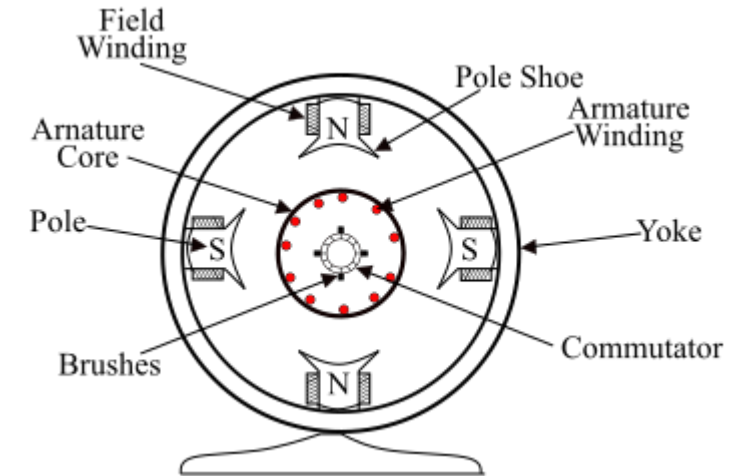


Construction of a DC Generator



Working of a DC Generator

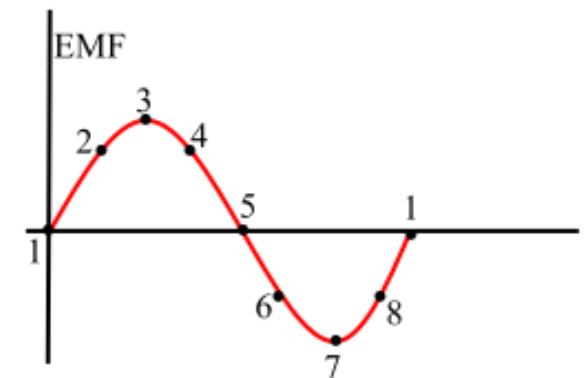
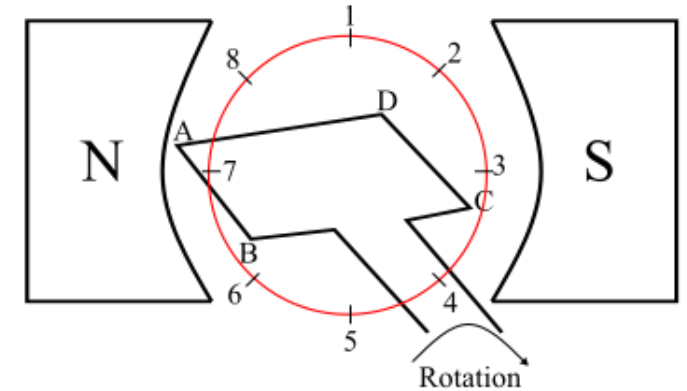
Consider a single loop DC generator (as shown in the figure), in this a single turn loop 'ABCD' is rotating clockwise in a uniform magnetic field with a constant speed. When the loop rotates, the magnetic flux linking the coil sides 'AB' and 'CD' changes continuously. This change in flux linkage induces an EMF in coil sides and the induced EMF in one coil side adds the induced EMF in the other.



Working of a DC Generator

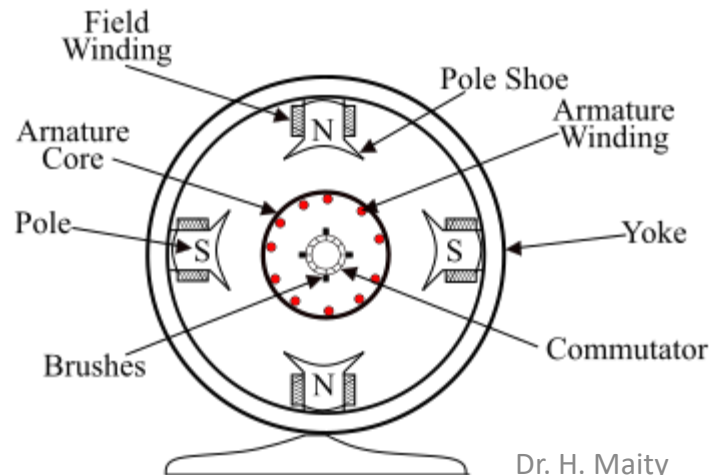
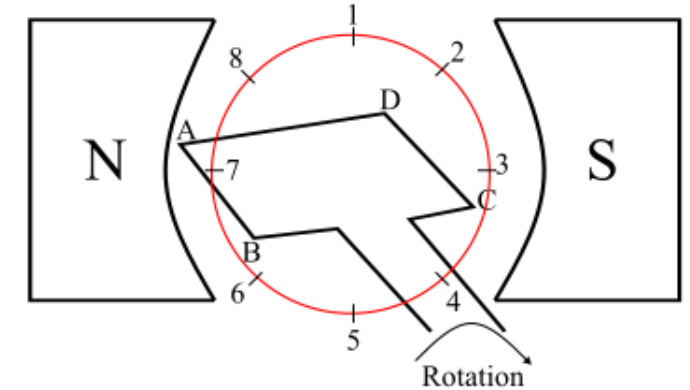
The EMF induced in a DC generator can be explained as follows

1. When the loop is in position-1, the generated EMF is zero because, the movement of coil sides is parallel to the magnetic flux.
2. When the loop is in position-2, the coil sides are moving at an angle to the magnetic flux and hence, a small EMF is generated.
3. When the loop is in position-3, the coil sides are moving at right angle to the magnetic flux, therefore the generated EMF is maximum.
4. When the loop is in position-4, the coil sides are cutting the magnetic flux at an angle, thus a reduced EMF is generated in the coil sides.
5. When the loop is in position-5, no flux linkage with the coil side and are moving parallel to the magnetic flux. Therefore, no EMF is generated in the coil.
6. At the position-6, the coil sides move under a pole of opposite polarity and hence the polarity of generated EMF is reversed. The maximum EMF will generate in this direction at position-7 and zero when at position-1. This cycle repeats with revolution of the coil.

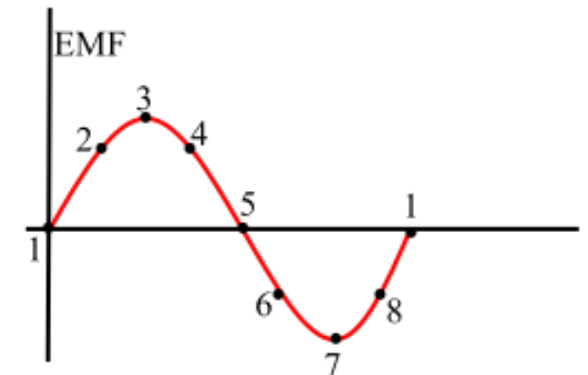


Working of a DC Generator

It is clear that the generated EMF in the loop is alternating one. It is because any coil side (say AB) has EMF in one direction when under the influence of N-pole and in the other direction when under the influence of S-pole. Hence, when a load is connected across the terminals of the generator, an alternating current will flow through it. Now, by using a commutator, this alternating emf generated in the loop can be converted into direct voltage.



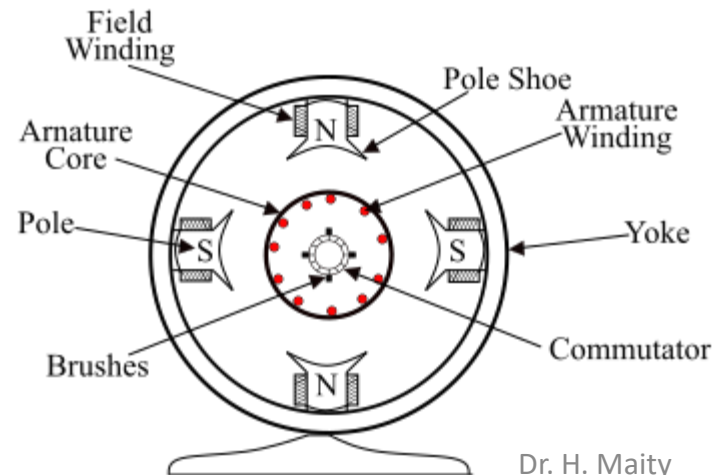
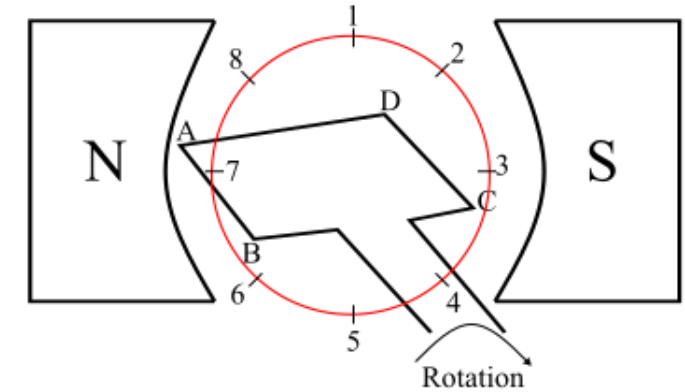
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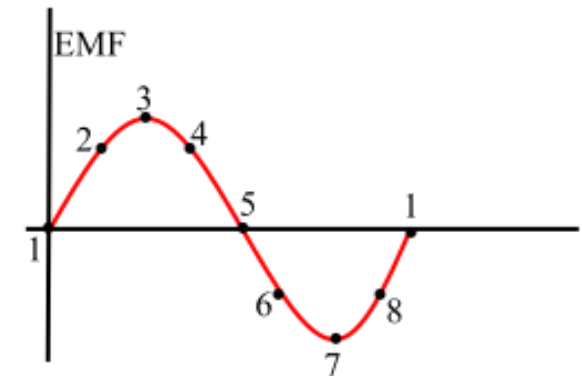
EMF Equation of a DC Generator

The derivation of **EMF equation for DC generator** has two parts:

1. Induced EMF of one conductor
2. Induced EMF of the generator



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EMF Equation of a DC Generator

Derivation for Induced EMF of One Armature Conductor

For one revolution of the conductor,

Let,

Φ = Flux produced by each pole in weber (Wb)

and

P = number of poles in the DC generator.

therefore,

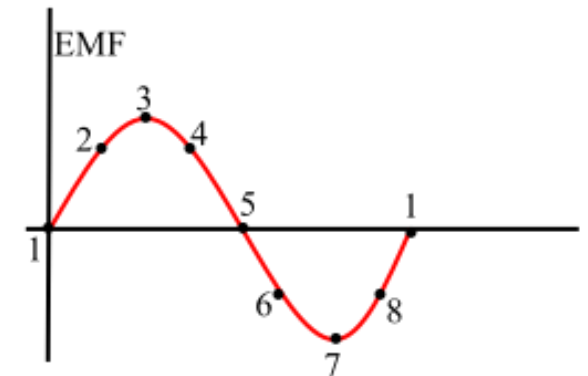
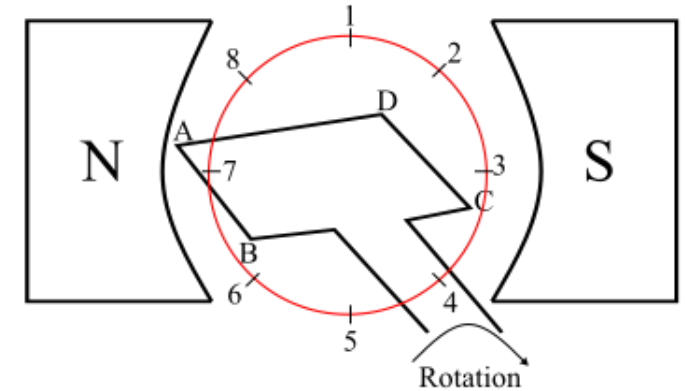
Total flux produced by all the poles = $\phi \times P$

And,

Time taken to complete one revolution = $\frac{60}{N}$

Where,

N = speed of the armature conductor in rpm.



EMF Equation of a DC Generator

Derivation for Induced EMF of One Armature Conductor

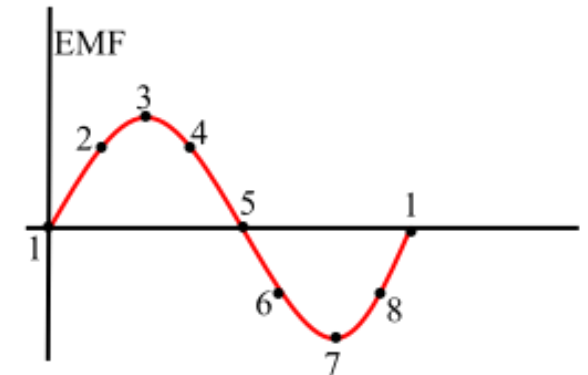
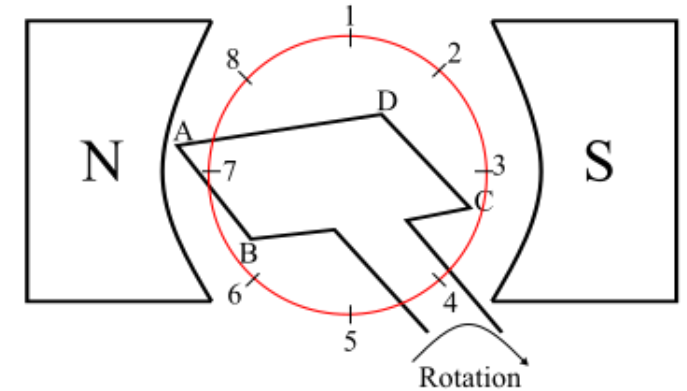
Now, according to Faraday's law of induction, the induced emf of the armature conductor is denoted by "e" which is equal to rate of cutting the flux.

$$e = \frac{d\phi}{dt} \text{ and } e = \frac{\text{total flux}}{\text{time take}}$$

Therefore,

Induced emf of one conductor is

$$e = \frac{\phi P}{\frac{60}{N}} = \phi P \frac{N}{60}$$



EMF Equation of a DC Generator

Derivation for Induced EMF for DC Generator

Let us suppose there are Z total numbers of conductor in a generator, and arranged in such a manner that all parallel paths are always in series.

Here,

Z = total numbers of conductor

A = number of parallel paths

Then,

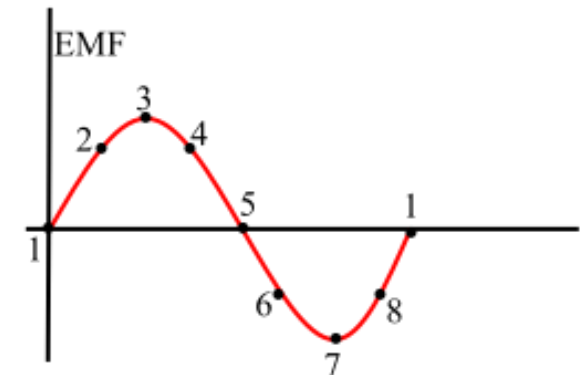
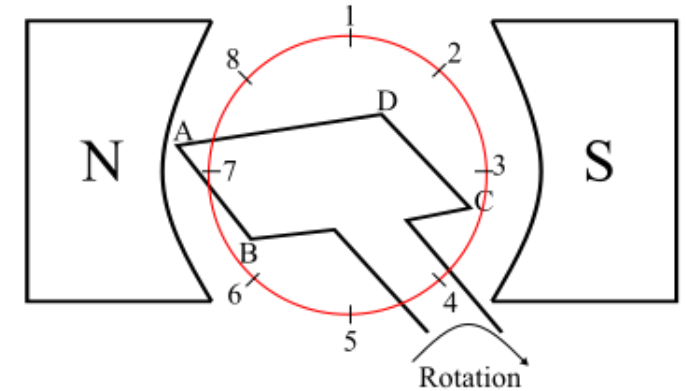
Z/A = number of conductors connected in series

We know that induced emf in each path is same across the line

Therefore,

Induced emf of DC generator

E = emf of one conductor $\times$ number of conductor connected in series.



EMF Equation of a DC Generator

Derivation for Induced EMF for DC Generator

Induced emf of DC generator

$E = \text{emf of one conductor} \times \text{number of conductor connected in series.}$

Induced emf of DC generator is

$$e = \phi P \frac{N}{60} \times \frac{Z}{A} \text{ volts}$$

Simple wave wound generator

Numbers of parallel paths are only $2 = A$

Therefore,

Induced emf for wave type of winding generator is

$$\frac{\phi P N}{60} \times \frac{Z}{2} = \frac{\phi Z P N}{120} \text{ volts}$$

For *Lap winding*, number of parallel paths $A = P$. Thus
For *Wave winding*, number of parallel paths $A = 2$. Thus

Simple lap-wound generator

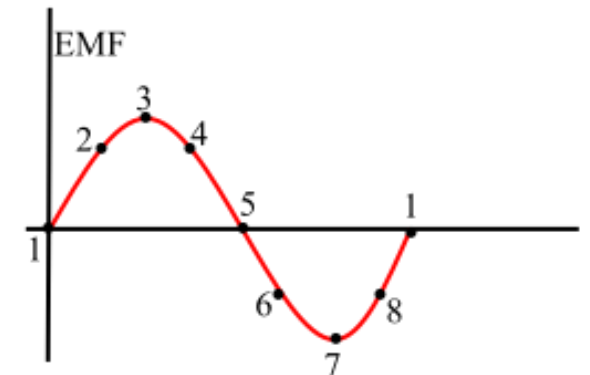
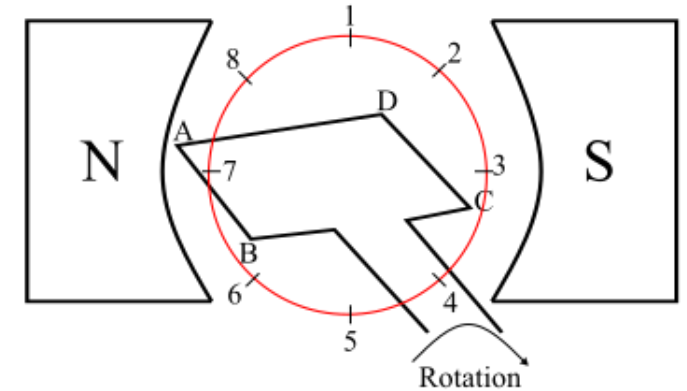
Here, number of parallel paths is equal to number of conductors in one path

i.e. $P = A$

Therefore,

Induced emf for lap-wound generator is

$$E = \frac{\phi N Z}{60}$$



EMF Equation of a DC Generator

Derivation for Induced EMF for DC Generator

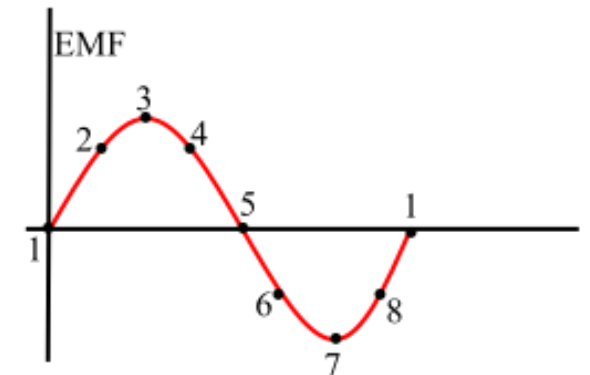
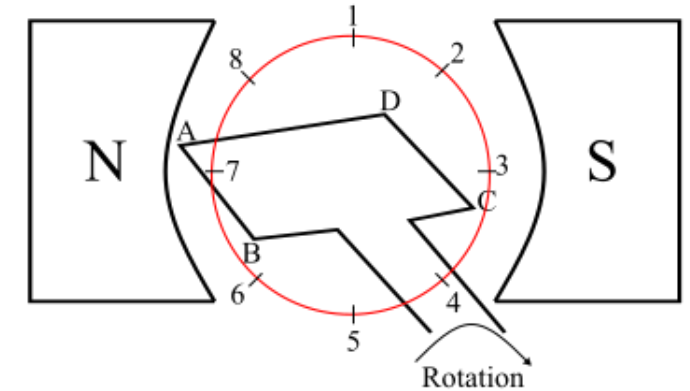
Induced emf of DC generator is

$$E = \frac{\phi P N Z}{60 A} \quad \text{e.m.f. equation}$$

$$E = \frac{\phi N Z}{60} \quad \text{for lap type as } A = P$$

$$E = \frac{\phi P N Z}{120} \quad \text{for wave type as } A = 2$$

For *Lap winding*, number of parallel paths $A = P$. Thus
For *Wave winding*, number of parallel paths $A = 2$. Thus



DC Generator

A 6-pole, DC generator has 800 conductors on its armature. The flux per pole is 0.035 Wb. The speed of rotation of the armature is 1500 RPM. Calculate the generated EMF when the armature is, (a) Lap wound, (b) Wave wound.

Solution:

(a). For Lap wound armature –

$$E_g = \frac{P\phi NZ}{60A} = \frac{6 \times 0.035 \times 1500 \times 800}{60 \times 6} = 700V$$

(b). For wave wound armature –

$$E_g = \frac{P\phi NZ}{60A} = \frac{6 \times 0.035 \times 1500 \times 800}{60 \times 2} = 2100V$$

DC Motor

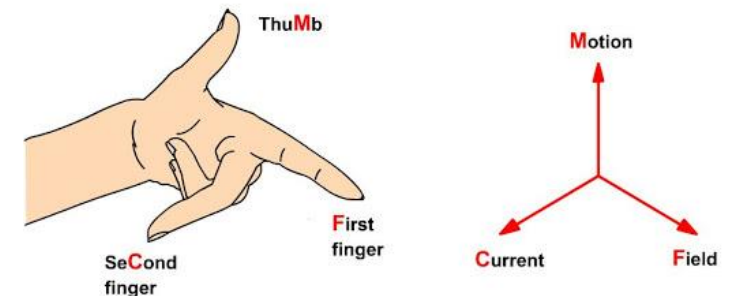
A DC motor is an electrical machine that converts electrical energy into mechanical energy. In a DC motor, the input electrical energy is the direct current which is transformed into the mechanical rotation.

Working Principle of DC Motor

A DC motor is an electrical machine which converts electrical energy into mechanical energy. The basic working principle of the DC motor is that whenever a current carrying conductor places in the magnetic field, it experiences a mechanical force.

Fleming's Left Hand Rule:

If we stretch the first finger, second finger and thumb of our left hand to be perpendicular to each other, and first finger represents the direction of the magnetic field, the second finger represents the direction of the current, then the thumb represents the direction of the force experienced by the current carrying conductor.



DC Motor

$$F = BIL \text{ Newtons}$$

Where,

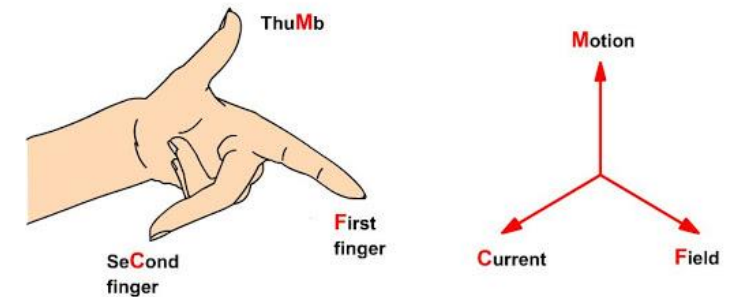
B = magnetic flux density,

I = current and

L = length of the conductor within the magnetic field.

When armature winding is connected to a DC supply, an electric current sets up in the winding. Permanent magnets or field winding (electromagnetism) provides the magnetic field. In this case, current carrying armature conductors experience a force due to the magnetic field, according to the principle stated above.

The Commutator is made segmented to achieve unidirectional torque. Otherwise, the direction of force would have reversed every time when the direction of movement of the conductor is reversed in the magnetic field. This is how a DC motor works!



DC Motor

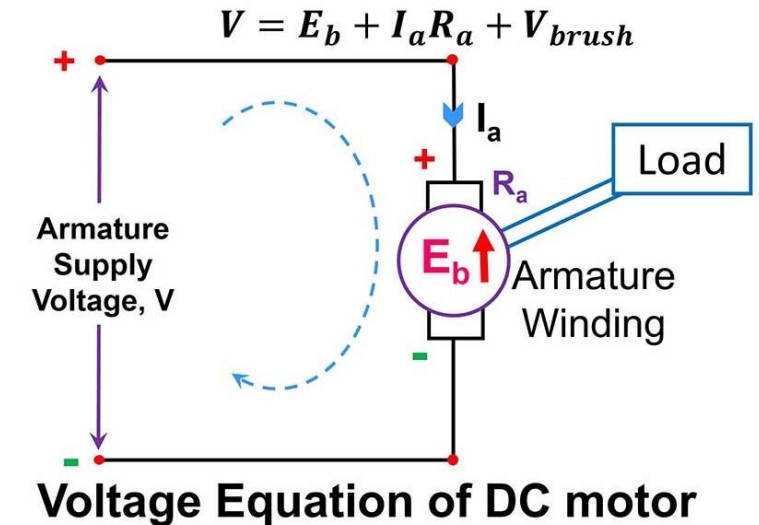
Back- EMF of DC motor

According to the fundamental law of nature, no energy conversion is possible until there is something to oppose the conversion. In case of generators, magnetic drag provides this opposition.

But in the case of dc motors, there is back emf. Presence of the back emf makes a dc motor 'self-regulating'.

When the armature of a motor is rotating, the conductors are also cutting the magnetic flux lines and hence according to the Faraday's law of electromagnetic induction, an emf induces in the armature conductors.

The direction of this induced emf is such that it opposes the armature current (I_a). The circuit diagram below illustrates the direction of the back emf and armature current.



$$E_b = \frac{\phi P N Z}{60 A}$$

DC Motor

Back- EMF of DC motor

Supply voltage = Back EMF + Voltage drop across armature + Voltage drop across Brushes

Now, mathematically express as,

$$\therefore V = E_b + I_a R_a + V_{brush}$$

But the voltage drop across brushes is negligible, so we can neglect this voltage drop.

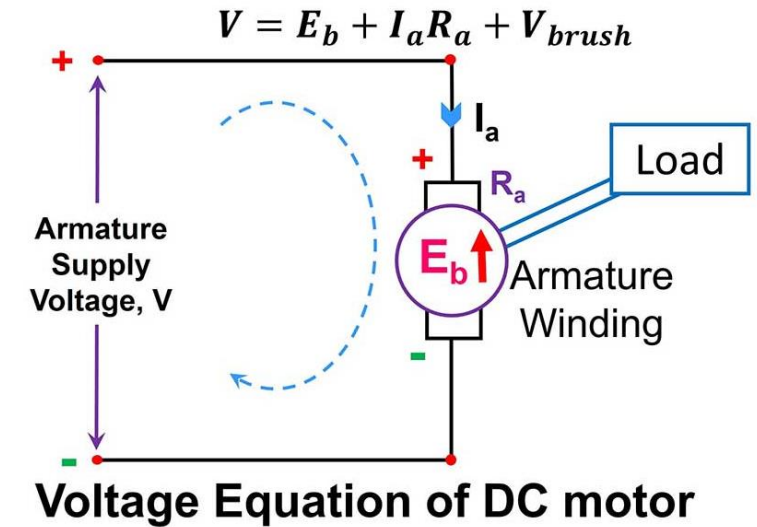
$$\therefore V = E_b + I_a R_a$$

This is the final expression for voltage and from above equation we can find out armature current (I_a) or back EMF by rearranging this equation.

Remember that the back emf E_b is always less than the armature voltage V . i.e. $E_b < V$, and the difference between V and E_b is very small under all the normal operating conditions.

The Back EMF control the armature current to its normal safe value.

When mechanical load increase on motor, motor speed down. But value of back EMF is proportional to Speed of motor, so back emf also decrease. Hence value of armature current increases.



$$E_b = \frac{\phi P N Z}{60 A}$$

DC Motor

Power Equation of a DC Motor:

Let us repeat the voltage Equation of dc motor,

$$\therefore V = E_b + I_a R_a$$

We know the basic equation of power i.e. $P = V \times I$.

So we can multiply both the sides of this equation by I_a to get,

$$\therefore V \times I_a = E_b \times I_a + I_a \times I_a R_a$$

$$\therefore VI_a = E_b I_a + I_a^2 R_a$$

In above power equation of dc motor VI_a is the electrical power supplied to the armature. On other side $E_b I_a$ is electrical equivalent of the mechanical power produce by the dc motor. And $I_a^2 R_a$ is the power loss in armature winding.

This is the final equation for power of DC motor.



DC Motor

Significance of Back E.M.F. in DC Motor

- **The presence of back emf makes the D.C. motor a *self-regulating machine*** i.e., it makes the motor to draw as much armature current as is just sufficient to develop the torque required by the load.

$$\text{Armature current (I}_a\text{), } I_a = \frac{V - E_b}{R_a}$$

- **When the motor is running on no load**, small torque is required to overcome the friction and windage losses. Therefore, the armature current I_a is small and the **back emf** is nearly equal to the applied voltage.
- **If the motor is suddenly loaded**, the first effect is to cause the armature to slow down. Therefore, the speed at which the armature conductors move through the field is reduced and hence the **back emf** E_b falls.
- The decreased **back emf** allows a larger current to flow through the armature and larger current means increased driving torque.
- Thus, the driving torque increases as the motor slows down. The motor will stop slowing down when the armature current is just sufficient to produce the increased torque required by the load.

DC Motor

Condition for Maximum Mechanical Power

P_m = The Mechanical power developed by the motor is

$$P_m = E_b I_a \quad V = E_b + I_a R_a$$

$$P_m = V I_a - I_a^2 R_a$$
$$\frac{dP_m}{dI_a} = V - 2I_a R_a = 0$$

$$\therefore I_a R_a = \frac{V}{2}$$

$$\text{As } V = E_b + I_a R_a \text{ and } I_a R_a = \frac{V}{2}$$

$$\therefore E_b = \frac{V}{2}$$

Applications of DC Motors

Shunt Motor Applications:

- Lathe Machine
- Drilling Machine
- Milling Machine
- Printing Machine
- Pumps
- Blowers
- Fans

Series Motor Applications:

- Electric trains,
- Diesel-electric locomotives,
- Cranes,
- Trolley cars and trolley buses,
- Conveyers etc.

Cumulative Compound Motor Applications:

- Elevators
- Rolling mills



Problem on DC Motor

A 250 V shunt motor takes a total current of 20 A. The shunt field and armature resistances are $200\ \Omega$ and $0.3\ \Omega$ respectively. Determine (i) value of Back E.M.F. and (ii) gross mechanical power in the armature.

Solution: ANS:

$$E_b = 244.4\text{ V}$$

$$\text{Mechanical Power developed} = 4582.5\text{ W.}$$



Problem on DC Motor

A 4 pole motor is fed at 440 V and takes an armature current of 50 A. The resistance of the armature circuit is 0.28Ω . The armature winding is wave-connected with 888 conductors and useful flux per pole is 0.023 Wb. Calculate the speed of the motor.

Solution: ANS:

$N=626$ r.p.m



Problem on DC Motor

The counter e.m.f. of a shunt motor is 227 V. The field resistance is $160\ \Omega$ and field current is 1.5 A. If the line current is 39.5 A. Find the armature resistance. Also find the armature current when the motor is stationary.

Solution: ANS:

$$R_a = 0.342\ \Omega$$

$$I_a = 38\ \text{A}$$

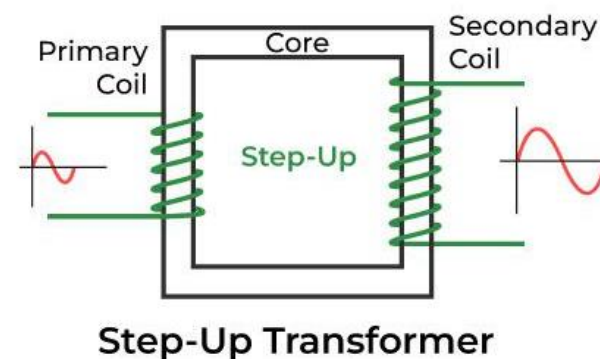
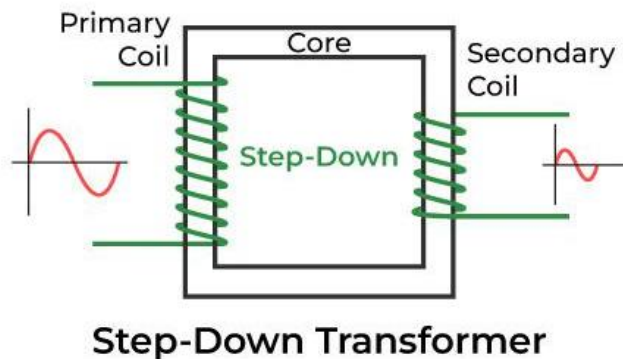
Single Phase Transformer

A transformer is a **static device that transfers electric energy from one alternating-current circuit to one or more other circuits**, either increasing (stepping up) or reducing (stepping down) the voltage.

Types of Transformer

Transformer types based on Voltage Level

There are primarily two types of Transformer based on the operating voltage. The following are some of them:



Single Phase Transformer

Types of Transformer

- **Step-down Transformer:** The primary voltage is converted to a lower voltage across the secondary output using a step-down transformer. The number of windings on the primary side of a step-down transformer is more than on the secondary side. As a result, the overall secondary-to-primary winding ratio will always be less than one. Step-down transformer are used in electrical systems that distribute electricity over long distances and operate at extremely high voltages to ensure minimum loss and economical solutions. Step-down transformer are used to change high-voltage into low-voltage supply lines.
- **Step-up Transformer:** The secondary voltage of a step-up transformer is raised from the low primary voltage. Because the primary winding has fewer turns than the secondary winding in this sort of transformer, the ratio of the primary to secondary winding will be greater than one. Step-up transformer are frequently used in electronics stabilizers, inverters, and other devices that convert low voltage to a significantly higher voltage. A step-up transformer is also used in the distribution of electrical power. For applications connected to power distribution, high voltage is necessary. In the grid, a step-up transformer is used to raise the voltage level prior to distribution.

Single Phase Transformer

Transformer Types based on Core Material

Different types of Transformer are used in the power and electronics industries, depending on the core materials, which are:

- **Iron Core Transformer:** Multiple soft iron plates are used as the core of an iron core transformer. The iron's strong magnetic properties of the iron core transformer have extremely high flux linkage. As a result, the iron core transformer has high efficiency. The soft iron core plates come in a variety of sizes and shapes. A few typical shapes include E, I, U, and L.
- **Ferrite Core Transformer:** Due to its high magnetic permeability, a ferrite core transformer uses one. In the high-frequency application, this kind of transformer provides incredibly low losses. In high-frequency applications like switch mode power supplies (SMPS), RF-related applications, etc., ferrite core transformer are used as a result.
- **Toroidal Core Transformer:** Iron core or ferrite core are two examples of toroid-shaped core materials used in transformer. For their excellent electrical performance, toroids, which have a ring- or donut-shaped core material, are frequently used. The ring form results in very low leakage inductance and extremely high inductance and Q factors.
- **Air Core transformer:** The core material of an air core transformer is not a real magnetic core. The air is used solely in the air-core transformer flux linkage. The primary coil of an air-core transformer generates an alternating current, producing an electromagnetic field all around it.



Single Phase Transformer

Transformer Types

Transformer types based on Voltage Level

Transformer Types based on Core Material

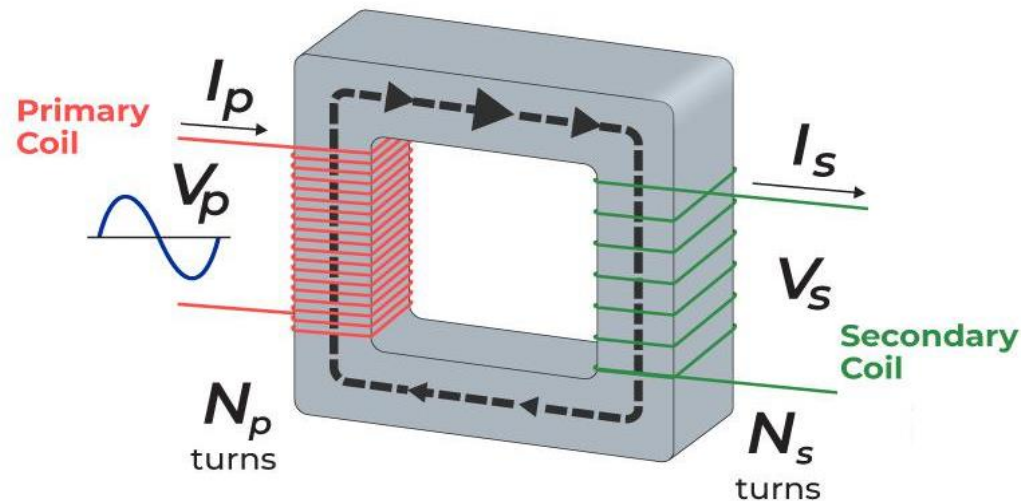
Types of Transformer based on Usage

Transformer Types based on Winding Arrangement

Single Phase Transformer

Working Principle of a Transformer

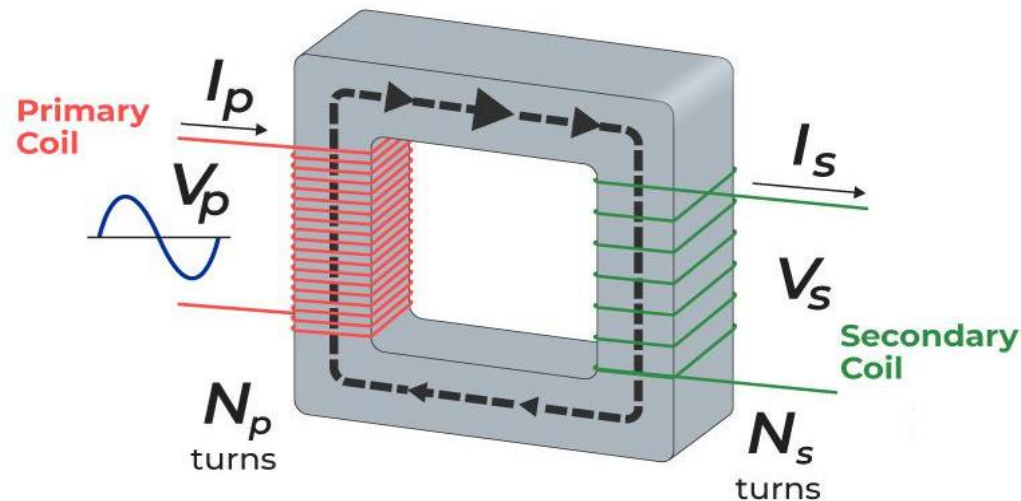
The fundamental principle of how the transformer functions are mutual induction between the two coils or Faraday's Law of Electromagnetic Induction. The laminated silicon steel core of the transformer is covered by two distinct windings. According to the diagram below, the primary winding is the one to which the AC supply is connected, and the secondary winding is the one to which the load is connected. Only alternating current can be used because mutual induction between the two windings requires an alternating flux.



Single Phase Transformer

Working Principle of a Transformer

According to Faraday's rule of electromagnetic induction, this alternating flux links the transformer primary and secondary windings magnetically and generates EMFs E_1 in the primary winding and E_2 in the secondary winding. The EMF (E_1) is referred to as the primary EMF, while the EMF (E_2) is the secondary EMF.



$$E_1 = -N_1 \frac{d\phi_m}{dt}$$

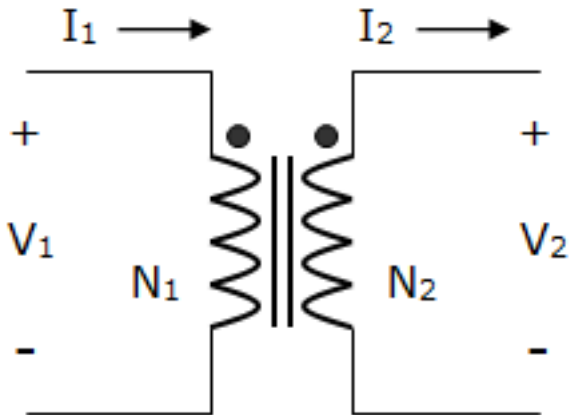
$$E_2 = -N_2 \frac{d\phi_m}{dt}$$

Dividing above equations, to obtain the ratio as:

$$\frac{E_1}{E_2} = \frac{N_1}{N_2}$$

Single Phase Transformer

$$\text{Turns Ratio} = \frac{N_1}{N_2} = a$$



$$a = \frac{N_1}{N_2} = \frac{V_1}{V_2} = \frac{I_2}{I_1}$$

$$P_1 = V_1 I_1 \quad P_2 = V_2 I_2$$

$$\text{Lossless Transformer: } P_1 = P_2$$

Single Phase Transformer

Efficiency of Transformer: The **Efficiency** of the transformer is defined as the ratio of useful output power to the input power. The input and output power are measured in the same unit. Its unit is either in Watts (W) or KW. **Transformer efficiency** is denoted by η .

$$\eta = \frac{\text{output power}}{\text{input power}} = \frac{\text{output power}}{\text{output power} + \text{losses}}$$

$$\eta = \frac{\text{output power}}{\text{output power} + \text{iron losses} + \text{copper losses}}$$

$$\eta = \frac{V_2 I_2 \cos \phi_2}{V_2 I_2 \cos \phi_2 + P_i + P_c}$$

Where,

- V_2 – Secondary terminal voltage
- I_2 – Full load secondary current
- $\cos \phi_2$ – power factor of the load
- P_i – Iron losses = hysteresis losses + eddy current losses
- P_c – Full load copper losses = $I_2^2 R_{es}$



Single Phase Transformer

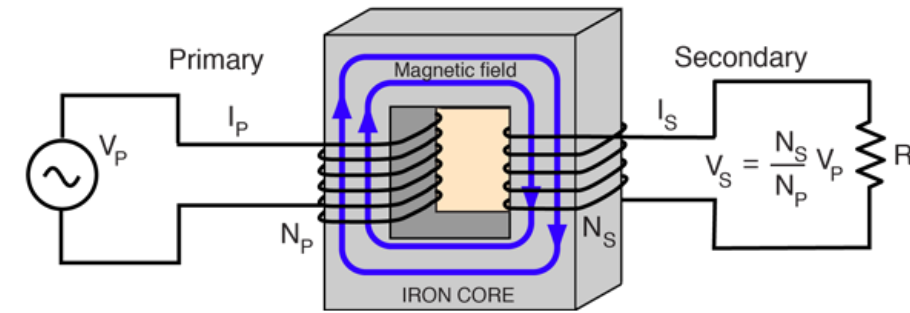
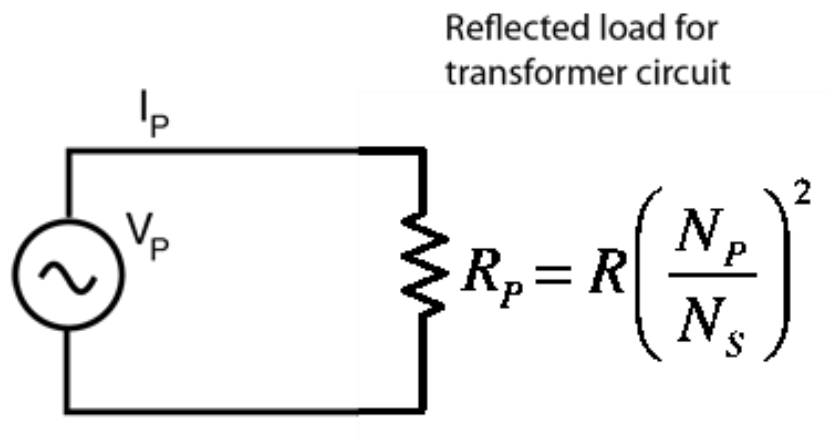
Ideal Transformer:

An ideal transformer is a purely theoretical transformer that has no losses at all, including no core losses, copper losses, or other transformer losses. This transformer is thought to be 100% efficient.

Single Phase Transformer

Reflected Load in a Transformer:

This **effective resistance as seen by the voltage source** is often called the "reflected load", evoking the image that the load resistor is "reflected" back across the transformer to be seen by the source. Once this reflected load is evaluated, the circuit can be considered like a simple resistive series circuit.



Ohm's law

$$R_P = \frac{V_P}{I_P}$$

Transformer relationship

$$V_P = V_S \frac{N_P}{N_S}$$

Power in = power out

$$I_P = \frac{I_S V_S}{V_P}$$

$$R_P = \frac{V_S \frac{N_P}{N_S}}{\frac{I_S V_S}{V_P}} = R \left(\frac{N_P}{N_S} \right) \left(\frac{V_P}{V_S} \right) = R \left(\frac{N_P}{N_S} \right)^2$$

Energy Losses in a Transformer

Energy losses do occur in actual transformer for the following reasons:

- **Flux Leakage:** Because some flux leaks from the core, not all flux generated by the primary coil make it to the secondary coil. This occurs as a result of the core's inadequate design or the presence of air holes in the core. It is possible to lower it by wrapping the primary and secondary coils over each other. It can also be lowered if the core is well-designed.
- **Windings Resistance:** Because the wire used for the windings has some electrical resistance, energy is wasted as a result of the heat generated in the windings. These are mitigated in high current, low voltage windings by utilizing thick wire with a high conductive substance.
- **Eddy Currents:** The alternating magnetic flux creates eddy currents in the iron core, resulting in energy losses through heating. By using a laminated core, the impact is decreased.
- **Hysteresis Loss:** In each AC cycle, the alternating magnetic field reverses the magnetization of the core. The loss of energy in the core occurs as heat owing to hysteresis loss, which is minimized by employing a magnetic material with a low hysteresis loss.



Applications of Transformer

The following are some of the most common uses for transformer:

1. Increasing or reducing the voltage level in an AC circuit to ensure the correct operation of the circuit's various electrical components.
2. It stops DC from flowing from one circuit to another.
3. It separates two separate electric circuits.
4. Before transmission and distribution can take place, the voltage level at the electric power plant must be increased.

Problem on Transformer

Example 1: A transformer has 1000 turns in the primary coil, and 8 A current flows through it. When the input power is 10 kW, and the output is 1000 V. Determine the number of turns in the secondary coil.

Sol:

Consider the case of an Ideal Transformer,

Given that, $P_{in} = P_{out} = 1000 \text{ W}$

But, $P_{out} = V_S I_S$

Now, the current through the secondary circuit is,

$$I_S = P_{out} / V_S = 10000 / 1000 = 10 \text{ A}$$

Therefore, the turns ratio of transformer is given by,

$$I_P / I_S = N_S / N_P$$

$$N_S = (I_P / I_S) N_P$$

$$= (8/10) \times 1000$$

$$= \mathbf{800 \text{ turns.}}$$



Problem on Transformer

Example 2: A transformer primary winding is powered by a 120 V ac source. If the turn ratio is 10, what does the secondary voltage equal?

Sol:

Problem on Transformer

Example 2: A transformer primary winding is powered by a 120 V ac source. If the turn ratio is 10, what does the secondary voltage equal?

Sol:

Given that, the turn ratio, $N_2/N_1 = 10$

And the voltage across the primary coil, $V_1 = 120$ V

Now, according to the transformer equation:

$$V_2/V_1 = N_2/N_1$$

Substituting the given values,

$$V_2/120 = 1/10$$

$$V_2 = \mathbf{12\ V}$$



Problem on Transformer

Example 3: The number of turns in the secondary coil of a 22 KVA, 2200V/220V single-phase transformer is 50, then find the number of primary turns. Neglect all kinds of losses in the transformer.

Sol:

Problem on Transformer

Example 3: The number of turns in the secondary coil of a 22 KVA, 2200V/220V single-phase transformer is 50, then find the number of primary turns. Neglect all kinds of losses in the transformer.

Sol:

The value of the turns ratio is

$$\begin{aligned} V_p/V_s &= 2200/220 \\ &= 10 = K \end{aligned}$$

Number of primary turns

The value of the primary turns can be determined as:-

$$N_p/N_s = K$$

$$N_p/50 = 10$$

$$N_p = \mathbf{500}$$

Transformer EMF Equation

As the magnetic flux varies sinusoidally, $\Phi = \Phi_{\max} \sin \omega t$, then the basic relationship between induced emf, (E) in a coil winding of N turns is given by:

emf = turns x rate of change

$$E = N \frac{d\Phi}{dt}$$

$$E = N \times \omega \times \Phi_{\max} \times \cos(\omega t)$$

$$E_{\max} = N \omega \Phi_{\max}$$

$$E_{\text{rms}} = \frac{N \omega}{\sqrt{2}} \times \Phi_{\max} = \frac{2\pi}{\sqrt{2}} \times f \times N \times \Phi_{\max}$$

$$\therefore E_{\text{rms}} = 4.44 f N \Phi_{\max}$$

•Where:

- f – is the flux frequency in Hertz, $= \omega/2\pi$
- N – is the number of coil windings.
- Φ – is the amount of flux in webers

This is known as the **Transformer EMF Equation**. For the primary winding emf, N will be the number of primary turns, (N_p) and for the secondary winding emf, N will be the number of secondary turns, (N_s).